

IGNASI ALIGUER-PIFERRER

Technology Leader | Innovation, Digital Transformation and High-Impact R&D

Personal Information

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Executive Summary

Startup co-founder and innovation leader with 15+ years of experience in technology commercialization, product development, and high-impact R&D in the Civil Engineering and Oil and Gas industries. Proven track record in raising seed startup funding (public and private), leading high-performance teams, and scaling innovative solutions for international markets. Skilled in translating complex technical challenges into market-ready solutions, fostering innovation culture, and driving strategic growth.

Core Competencies

- Innovation and go-to-market Strategies
 - SaaS and Cloud-Based Technology Development
 - Cross-Functional Leadership and Team Building
 - R&D and IP Portfolio Management
 - Emerging Tech Applications (AI, IoT, Predictive Analytics)
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Career Impact

- Raised \$6M in seed funding, including \$1.5M from Corporate VC and \$2.4M from EU EIC Accelerator program, for SAALG Geomechanics.
 - Launched a cloud-based SaaS product for predictive geomechanics (Daarwin), leading a 15-member engineering team using an Agile approach.
 - Launched 5 new products developed in-house (AI-based, predictive ML, iPhone App.)
 - Invented patented methods for Oil and Gas geomechanics (stress modelling and hydraulic fracture simulation).
 - Built novel pipelines and processes integrating numerical simulation, AI-driven analytics, and visualization tools, accelerating product development and client adoption.
 - Spanish Representative at the TC206 Committee - Observational Method (ISSMGE).
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Professional Experience

SAALG Geomechanics – Barcelona, Spain

Co-Founder and CTO | Chairman BoD | Sept 2016 – Present

- Defined the business model, product-market fit, go-to-market strategy, and roadmap for cloud-based SaaS product Daarwin.
- Led 15 engineers to develop advanced numerical modeling and real-time predictive tools for civil engineering and mining applications .
- Raised \$6M in funding (up to seed stage), including public funding (national and EU R&D programs) and corporate VC contributions.
- Closed strategic projects in multiple international markets (Germany, UAE, Guatemala, Colombia, Brasil) increasing global footprint and revenue potential.
- Established and led company's Board of Directors and its communications.

PetraBytes Corp. – Houston, TX, USA

Research Development Engineer | Jan 2015 – Aug 2016

- Developed Big Data and 3D visualization tools for Oil and Gas geomechanics.
- Built Finite Element pre- and post-processors for 3D geomechanics simulations.
- Optimized storage and visualization of complex reservoir models.
- Led client feedback sessions to prioritize feature development.

Polytechnical University of Catalonia (UPC) – Barcelona, Spain

Research Engineer | Dec 2012 – Dec 2014

- Developed patented technology for hydraulic fracture simulations in porous media in collaboration with Repsol.
- Created novel stress initialization techniques for geological structures.

Polytechnical University of Catalonia (UPC) – Barcelona, Spain

Research Associate | Apr 2010 – Dec 2012

- Formulated viscoplastic constitutive laws for rock interfaces/fractures.
- Collaborated on Finite Element research code DRAC for advanced geomechanics analysis.

University of Colorado at Boulder – Boulder, CO, USA

Visiting Scholar | Feb 2011 – Mar 2011, Dec 2011 – Jun 2012

- Collaborated on rock interface constitutive modeling.
- Speaker at a seminar at the US Bureau of Reclamation (Denver, CO).

Polytechnical University of Catalonia (UPC) – Barcelona, Spain

Graduate Research Assistant | Sept 2008 – Mar 2010

- Performed stability analysis of Canellas Dam abutment; developed procedures for safety evaluation using Finite Elements analysis.
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Education and Executive Development

Acceleration Programs

- iTwin Activate (Bentley Systems) – Exton, PA, USA | 2023
- Open Innovation Challenge (ACCIONA) – Madrid, Spain | 2022
- StartAD (NYU) – Abu Dhabi, UAE | 2020
- Lanzadera (Mercadona) – Valencia, Spain | 2020
- Mindbox (Deutsche Bahn) – Berlin, Germany | 2018
- The Collider (Mobile World Capital) – Barcelona, Spain | 2017

MSc & BSc in Geological Engineering

Polytech. Univ. of Catalonia (UPC) and Univ. de Barcelona (UB) | 2003 – 2008

- 1st Ranked GPA Award
 - Minor Thesis: *Biomechanical Analysis of an Intervertebral Prosthesis Using FEM*
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Technical Skills

- **Technology. Stack:** Python, Fortran, SQL noSQL DB, AWS (EC2, RDS, S3, DynamoDB), Git (Atlassian suite: JIRA, BitBucket, CD/CI)
 - **Engineering Tools:** Matlab, AutoCAD, Plaxis, GiD
 - **Languages:** Catalan and Spanish (Native), English (near-Native), French (Intermediate)
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Patents and Publications

- **WO2015107079A1** – Method to generate the in-situ state of stress in a domain Ω in a geological structure
 - **WO2015124633A1** – Method implemented in a computer for the numerical simulation of a porous medium
 - **Aliguer I., Carol I., and Sture S.** *Stress-driven integration strategies and m -AGC tangent operator for Perzyna viscoplasticity and viscoplastic relaxation: application to geomechanical interfaces.* Intl. J. Num. Anal. Meth. Geomech. (2017): 918-939
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Awards

- Santander X Spain Award | 2023
- Ground Engineering Awards: Digital Innovation and Editor's Award | 2023
- Forbes Spain top 100 business innovator | 2021
- Caixabank Emprendedor XXI | 2020
- DeepTech 4 Good Barcelona Industry 4.0 Award | 2019
- Expansion Startup Awards | 2019